ENGINE 2.4L (Continued)

(29) Disconnect crankshaft position sensor.
(30) Disconnect the engine block heater power cable, if equipped.
(31) Disconnect the front propshaft at the front differential and secure out of way.
(32) Remove the starter.
(33) Remove the ground straps from the engine
(34) Disconnect the exhaust pipes at the manifold.
(35) Remove the structural cover, if equipped.
(36) Remove torque convertor bolts, and mark location for reassembly.
(37) Remove transmission bellhousing to engine bolts.
(38) Loosen left and right engine mount thru bolts.

NOTE: It is not necessary to completely remove engine mount thru bolts, for engine removal.

(39) Lower the vehicle.
(40) Support the transmission with a suitable jack.
(41) Connect a suitable engine hoist to the engine.

CAUTION: The 2.4L engine with manual transmissions, can be removed without removing the manual transmission. Use caution when attempting this procedure as the clearance is tight.

(42) Remove engine from vehicle.

INSTALLATION - ENGINE ASSEMBLY

(1) Position the engine in the vehicle.

CAUTION: Use caution when installing 2.4L engine into vehicle equipped with manual transmission, as clearance is tight.

(2) Install both left and right side engine mounts into the frame mounts.
(3) Raise the vehicle.
(4) Install the transmission bellhousing to engine mounting bolts. Tighten the bolts to 41 N·m (30 ft. lbs.).
(5) Tighten the engine mount thru bolts.
(6) Install the torque convertor bolts.
(7) Connect the ground straps on the left and right side of the engine.
(8) Install the starter.
(9) Connect the crankshaft position sensor.
(10) Install the engine block heater power cable, if equipped.

CAUTION: The structural cover requires a specific torque sequence. Failure to follow this sequence may cause severe damage to the cover.

(11) Install the structural cover.
(12) Install the exhaust pipe.
(13) Connect the oxygen sensors.
(14) Lower vehicle.
(15) Connect the knock sensors.
(16) Connect the engine to body ground straps.
(17) Install the power brake booster vacuum hose.
(18) Install the breather hoses.
(19) Install the PCV hose.
(20) Install the fuel rail.
(21) Install the coil over plugs.
(22) Reconnect the engine wiring harness at the following points:

- Intake air temperature (IAT) sensor
- Fuel Injectors
- Throttle Position (TPS) Switch
- Idle Air Control (IAC) Motor
- Engine Oil Pressure Switch
- Engine Coolant Temperature (ECT) Sensor
- Manifold Absolute Pressure MAP) Sensor
- Camshaft Position (CMP) Sensor
- Coil Over Plugs
- Crankshaft Position Sensor

(23) Connect lower radiator hose.
(24) Connect upper radiator hose.
(25) Connect throttle and speed control cables.
(26) Install the heater hose assembly.
(27) Install coolant recovery bottle.
(28) Install the power steering pump.
(29) Install the generator.
(30) Install the A/C compressor.
(31) Install the drive belt.
(32) Install the fan shroud with the electric fan assembly.
(33) Install the radiator core support bracket.
(34) Install the air cleaner assembly.
(35) Refill the engine cooling system.
(36) Install the hood.
(37) Check and fill engine oil.
(38) Connect the battery negative cable.
(39) Start the engine and check for leaks.